Instagram Engagement Behaviour Analysis

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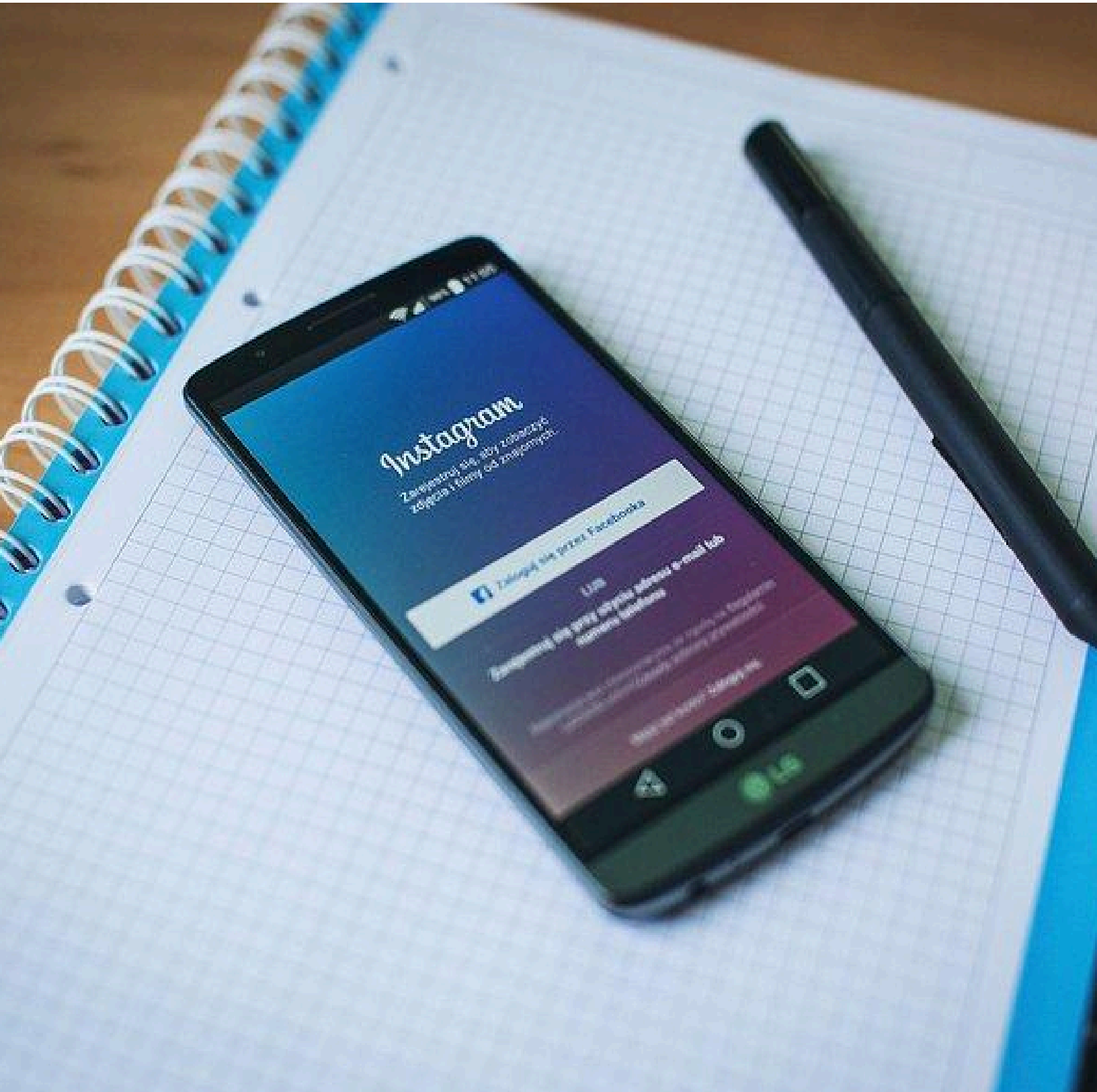

1. Introduction

Social media platforms generate large amounts of behavioural data that can be analyzed to understand user engagement patterns. This case study analyzes my personal Instagram interaction data to identify patterns in engagement, activity timing, and account interaction behaviour.

The dataset was obtained from the Instagram data export feature and includes records of liked posts, story likes, followers, and following activity. The objective of this project is to analyze how interactions occur over time and identify behavioural insights from the engagement data.

2. Business Task

The goal of this analysis is to understand **Instagram engagement behaviour patterns** by answering the following questions:

- What type of interactions occur most frequently?
- Which accounts receive the most engagement?
- When is Instagram activity highest during the day?
- Which days show the highest interaction levels?
- How has engagement changed over time?
- What patterns exist in followers and following activity?

3. Data Source

The dataset used for this analysis was obtained through the Instagram data export feature. The exported files contained interaction records in CSV format, including:

- liked posts

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- story likes
- followers list
- following list

Each dataset contained the following key fields:

Column	Description
account_name	Instagram account interacted with
like_datetime	Timestamp of interaction
year	Year of interaction
month	Month of interaction
day	Day of week
hour	Hour of interaction
interaction_type	Type of interaction

These datasets were consolidated into a single analytical dataset to enable integrated analysis.

4. Tools Used

The following tools were used to complete the project:

- **Google Sheets** – data cleaning and consolidation
- **MySQL** – data querying and analysis
- **Tableau Public** – interactive data visualization and dashboard creation

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5. Data Cleaning and Preparation

The raw exported data contained multiple CSV files with separate interaction records. The following steps were taken to prepare the dataset:

1. Removed repeated header rows and blank rows from the datasets.
2. Standardized column names across all datasets.
3. Added a new column **interaction_type** to distinguish between:
    - post likes
    - story likes
    - followers
    - following
4. Extracted **year, month, day, and hour** from timestamp values.
5. Consolidated all interaction datasets into a single master dataset.
6. Verified the dataset for missing values and formatting inconsistencies.

The cleaned dataset was then prepared for analysis and visualization.

6. Data Analysis

SQL queries were used to explore engagement patterns across different dimensions. The analysis focused on identifying behavioural insights related to:

- interaction frequency
- most engaged accounts
- time-based activity patterns

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- engagement trends over time
- follower and following growth

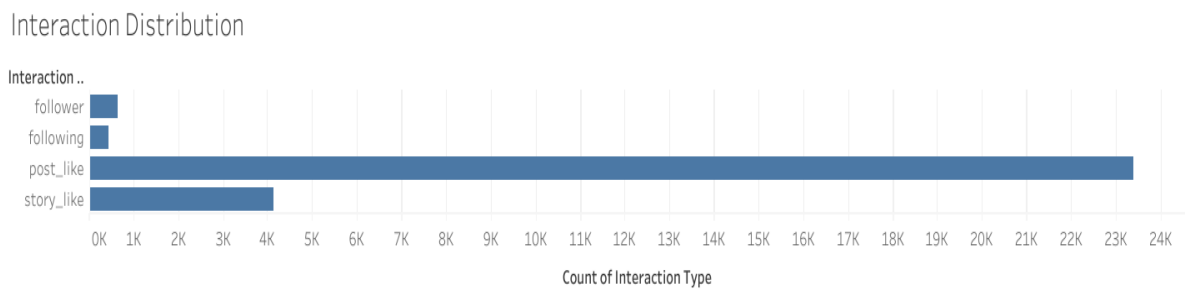
Aggregations and grouping techniques were used to calculate counts and trends across the dataset.

7. Data Visualization

Interactive dashboards were developed using **Tableau Public** to present the analytical insights. The visualizations created include:

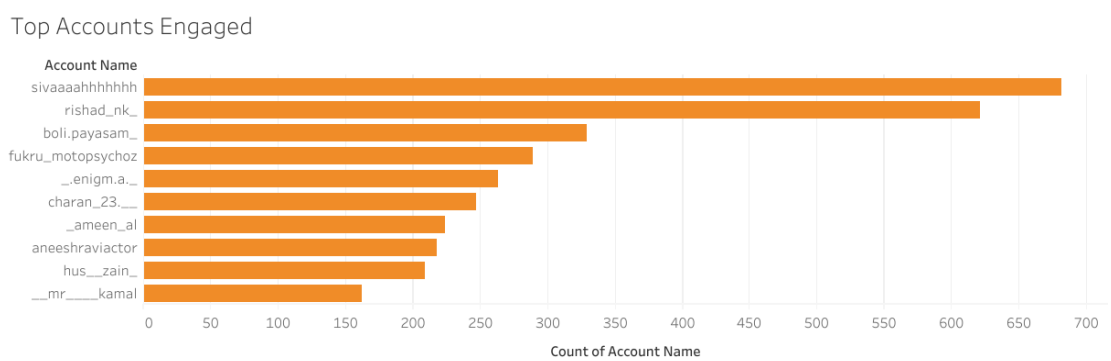
- **Interaction Distribution**

This visualization shows the distribution of different interaction types including liked posts, story likes, followers, and following activity.



- **Top Accounts Engaged**

This chart identifies the accounts that receive the highest number of interactions.



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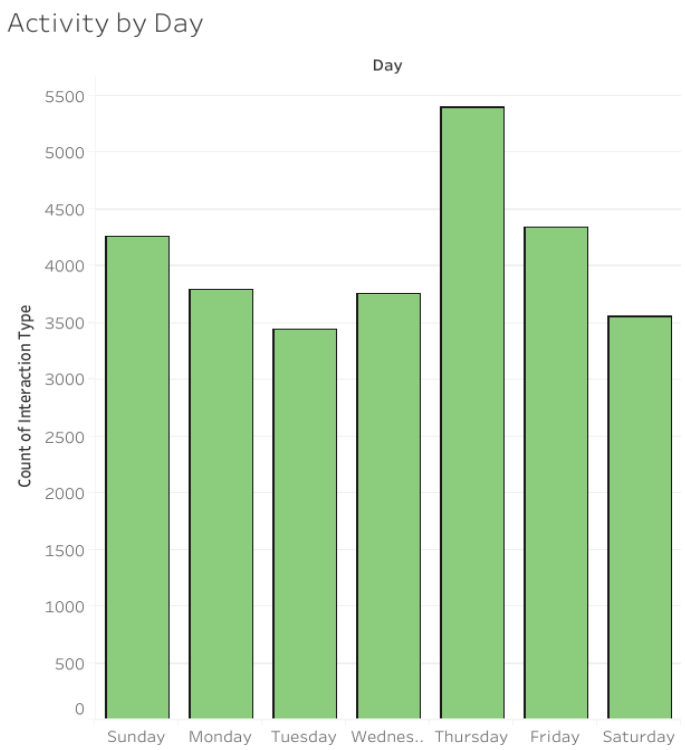
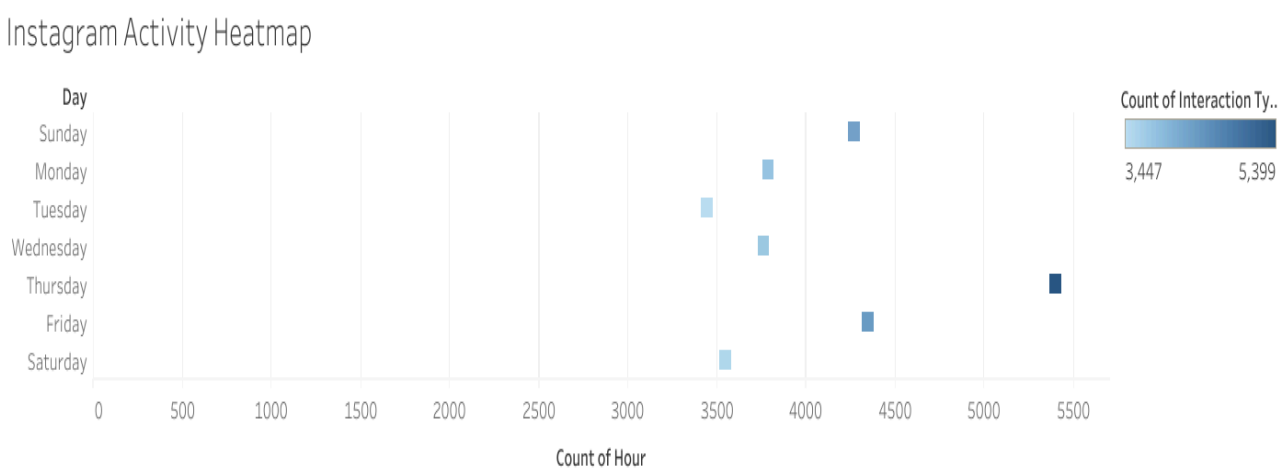
- **Activity by Day**

This visualization highlights the days of the week with the highest engagement activity.



- **Instagram Activity Heatmap**

The heatmap shows engagement intensity across hours of the day and days of the week, revealing peak usage periods.



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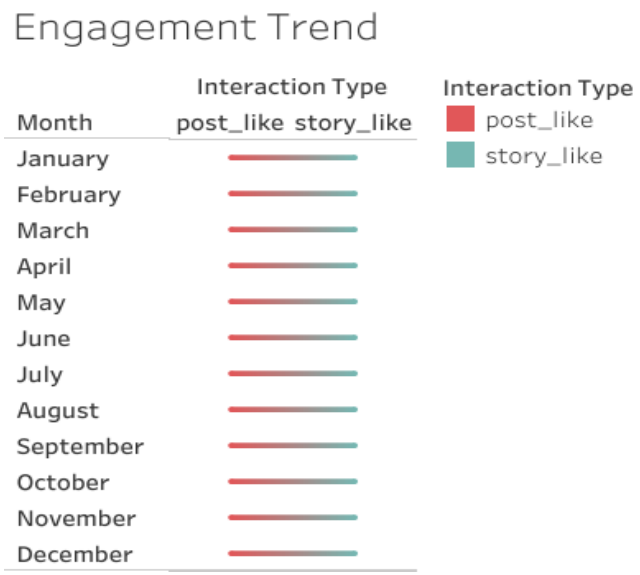
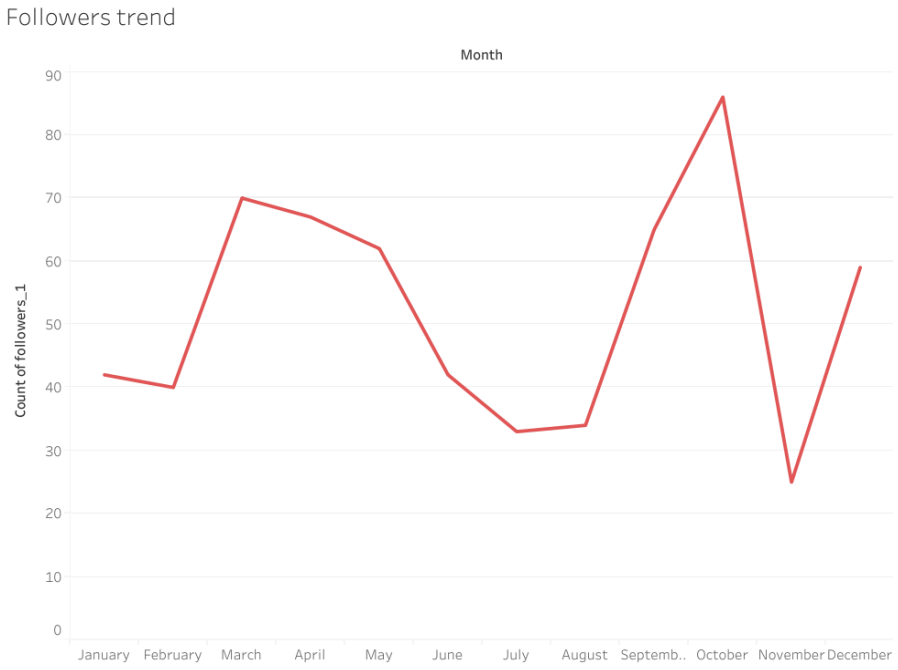
- **Engagement Trend**

This chart shows how engagement has changed over time, helping identify behavioural trends.



- **Followers Trend**

This visualization shows the pattern of follower additions over time.



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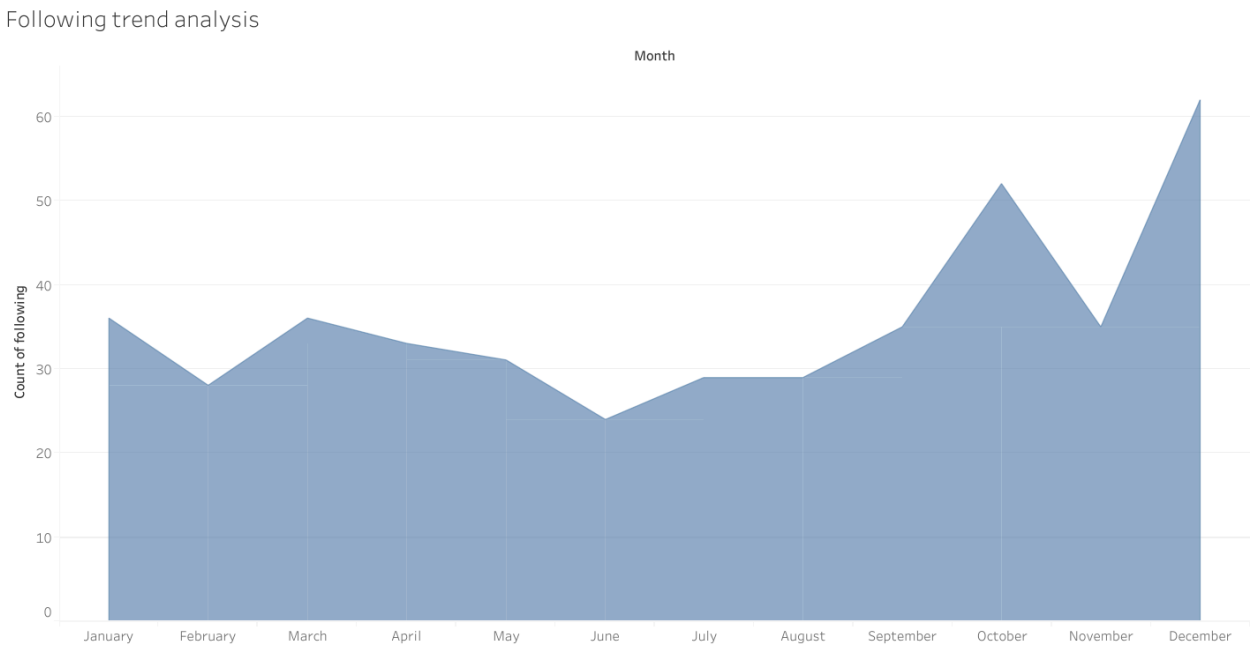
- **Following Trend Analysis**

This chart analyzes patterns in accounts followed over time.



8. Key Insights

The analysis revealed several behavioural insights:

1. Post likes represent the most frequent type of interaction.

2. Engagement tends to peak during specific hours of the day, particularly in the evening.

3. Activity patterns vary across weekdays and weekends.

4. Certain accounts receive significantly higher interaction levels compared to others.

5. Engagement trends show fluctuations across different time periods.

These findings highlight how personal engagement behaviour changes depending on time and interaction type.

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9. Conclusion

This project demonstrates how personal social media interaction data can be analyzed to uncover behavioural patterns. By cleaning, consolidating, and analyzing the dataset, it was possible to identify trends in engagement timing, interaction types, and account engagement levels.

The project also demonstrates practical data analytics skills including:

- data cleaning
- dataset consolidation
- SQL-based analysis
- data visualization
- insight generation

The resulting dashboard provides an interactive way to explore Instagram engagement behaviour and better understand user interaction patterns.

10. Author

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Data Analytics Portfolio Project

Tools Used:
Google Sheets | MySQL | Tableau Public

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